

Let $K \geq 2$ and let $\nu_1, \dots, \nu_K$ be probability distributions on $[0, 1]$, with respective means $\mu_1, \dots, \mu_K$. We write

$$\mu^* = \max_{1 \leq a \leq K} \mu_a, \quad \Delta_a = \mu^* - \mu_a, \quad a = 1, \dots, K. \quad (1)$$

Let $T \geq K$. At each round $t = 1, \dots, T$, an agent chooses an arm $A_t \in \{1, \dots, K\}$ based on the history $H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1})$ and, possibly, on an independent random draw, then receives a reward X_t such that, conditionally on (H_{t-1}, A_t) , X_t has distribution ν_{A_t} . For $a \in \{1, \dots, K\}$ and $t \in \{1, \dots, T\}$, we set

$$N_a(t) = \sum_{s=1}^t \mathbf{1}\{A_s = a\}, \quad \hat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^t X_s \mathbf{1}\{A_s = a\} \quad \text{if } N_a(t) \geq 1, \quad (2)$$

and we define the regret of the agent as

$$R_T = T\mu^* - \mathbb{E} \left[\sum_{t=1}^T X_t \right]. \quad (3)$$

Exercise 1 (Regret decomposition). In this exercise, the policy of the agent is arbitrary unless stated otherwise.

1. Let $t \in \{1, \dots, T\}$. Show that $\mathbb{E}[X_t] = \mathbb{E}[\mu_{A_t}]$.
Hint: write $\mathbb{E}[X_t] = \mathbb{E}[\mathbb{E}[X_t | H_{t-1}, A_t]]$.
2. Show that $\sum_{t=1}^T \mu_{A_t} = \sum_{a=1}^K \mu_a N_a(T)$, then that

$$R_T = \sum_{a=1}^K \Delta_a \mathbb{E}[N_a(T)]. \quad (4)$$

3. Assume that the A_t are i.i.d., uniform on $\{1, \dots, K\}$ and independent of the rewards (uniform exploration). Compute $\mathbb{E}[N_a(T)]$ for any a , then R_T .
4. Let $\varepsilon \in (0, 1]$. Consider an agent that, at each round t , draws with probability ε (independently of H_{t-1}) an arm uniformly on $\{1, \dots, K\}$, and otherwise chooses A_t according to any rule based on H_{t-1} (ε -greedy). Show that $\mathbb{P}(A_t = a) \geq \varepsilon/K$ for any a and t , and deduce that

$$R_T \geq \frac{\varepsilon T}{K} \sum_{a=1}^K \Delta_a.$$

Deduce that if $\Delta_a > 0$ for at least one arm a , then R_T/T does not tend to 0 as $T \rightarrow \infty$.

Exercise 2 (Greedy can lock onto the wrong arm). Assume $K = 2$, $\nu_1 = \mathcal{B}(p)$ and $\nu_2 = \mathcal{B}(q)$ with $0 < q < p < 1$, and $T \geq 3$. The greedy strategy plays $A_1 = 1$, $A_2 = 2$, then, for $t \geq 3$,

$$A_t = \begin{cases} 1 & \text{if } \hat{\mu}_1(t-1) \geq \hat{\mu}_2(t-1) \\ 2 & \text{otherwise} \end{cases}. \quad (5)$$

We write $E = \{X_1 = 0, X_2 = 1\}$.

1. Show that, on the event E , $A_t = 2$ for any $t \in \{2, \dots, T\}$.
Hint: proceed by induction on t and bound $\hat{\mu}_2(t-1)$ from below.
2. Compute $\mathbb{P}(E)$.
3. Deduce that $R_T \geq (1-p)q(p-q)(T-1)$. Compute this lower bound for $p = 0.6$, $q = 0.4$ and $T = 5000$.
4. Show that $N_1(T) = 1$ if and only if E occurs.

Exercise 3 (Explore-then-commit). Assume $K = 2$ and $\mu_1 > \mu_2$, and write $\Delta = \mu_1 - \mu_2$. Let $m \geq 1$ be an integer such that $2m \leq T$. The strategy ETC(m) alternates between the arms during the first $2m$ rounds ($A_{2i-1} = 1$ and $A_{2i} = 2$ for $i = 1, \dots, m$), then plays $A_t = \hat{a}$ for any $t > 2m$, where $\hat{a} = 1$ if $\hat{\mu}_1(2m) \geq \hat{\mu}_2(2m)$ and $\hat{a} = 2$ otherwise. We denote by $Y_{a,1}, \dots, Y_{a,m}$ the rewards collected on arm a during the first $2m$ rounds, and we admit that $Y_{1,1}, \dots, Y_{1,m}, Y_{2,1}, \dots, Y_{2,m}$ are independent, each $Y_{a,i}$ having distribution ν_a . We also admit Hoeffding's inequality: if $Z_1, \dots, Z_m$ are i.i.d. with values in $[\alpha, \beta]$ and expectation ζ , then, for any $\varepsilon > 0$,

$$\mathbb{P}\left(\frac{1}{m} \sum_{i=1}^m Z_i - \zeta \geq \varepsilon\right) \leq \exp\left(-\frac{2m\varepsilon^2}{(\beta - \alpha)^2}\right). \quad (6)$$

1. Show that $\mathbb{E}[N_2(T)] = m + (T - 2m)\mathbb{P}(\hat{a} = 2)$, and deduce that $R_T \leq m\Delta + T\Delta\mathbb{P}(\hat{a} = 2)$.
2. Let $Z_i = Y_{2,i} - Y_{1,i}$ for $i = 1, \dots, m$. Show that $\hat{\mu}_2(2m) - \hat{\mu}_1(2m) = \frac{1}{m} \sum_{i=1}^m Z_i$, then, using (6), that $\mathbb{P}(\hat{a} = 2) \leq \exp(-m\Delta^2/2)$.
3. Deduce that

$$R_T \leq g(m), \quad \text{where} \quad g(x) = x\Delta + T\Delta \exp(-x\Delta^2/2), \quad x > 0. \quad (7)$$

4. Assume $T\Delta^2 > 2$. Show that g attains its minimum on $(0, \infty)$ at

$$x^* = \frac{2}{\Delta^2} \log\left(\frac{T\Delta^2}{2}\right), \quad \text{and that} \quad g(x^*) = \frac{2}{\Delta} \left(1 + \log\left(\frac{T\Delta^2}{2}\right)\right).$$

5. Let $T = 5000$ and $\Delta = 0.2$. Compute approximate values of x^* and $g(x^*)$, and compare $g(x^*)$ with the regret of uniform exploration (exercise 1, question 3).
Hint: $\log 100 \approx 4.61$.